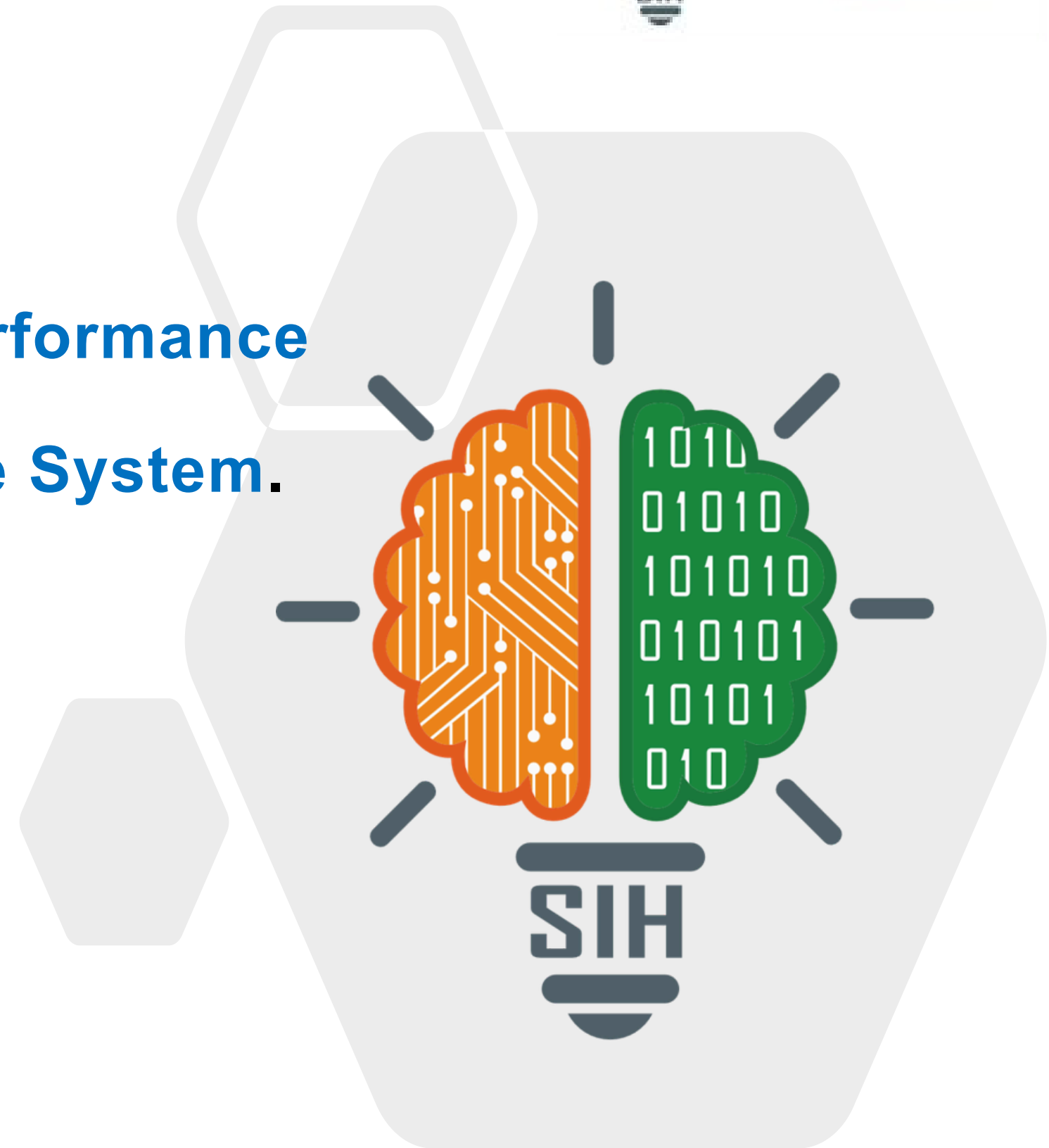


SMART INDIA HACKATHON 2026



- Problem Statement ID – **SIH26050**
- Problem Statement Title- **High Altitude Performance Optimization and Robust Design of Anti-Drone System.**
- Theme- **Robotics and Drones**
- PS Category- **Hardware**
- Team ID- **177186**
- Team Name - **Naan-Sense**





PROPOSED SOLUTION MULTI-MODAL SENSING

Radar + EO + LWIR + Passive RF
FMCW/mmWave Radar • EO Camera
• LWIR Thermal • Passive RF

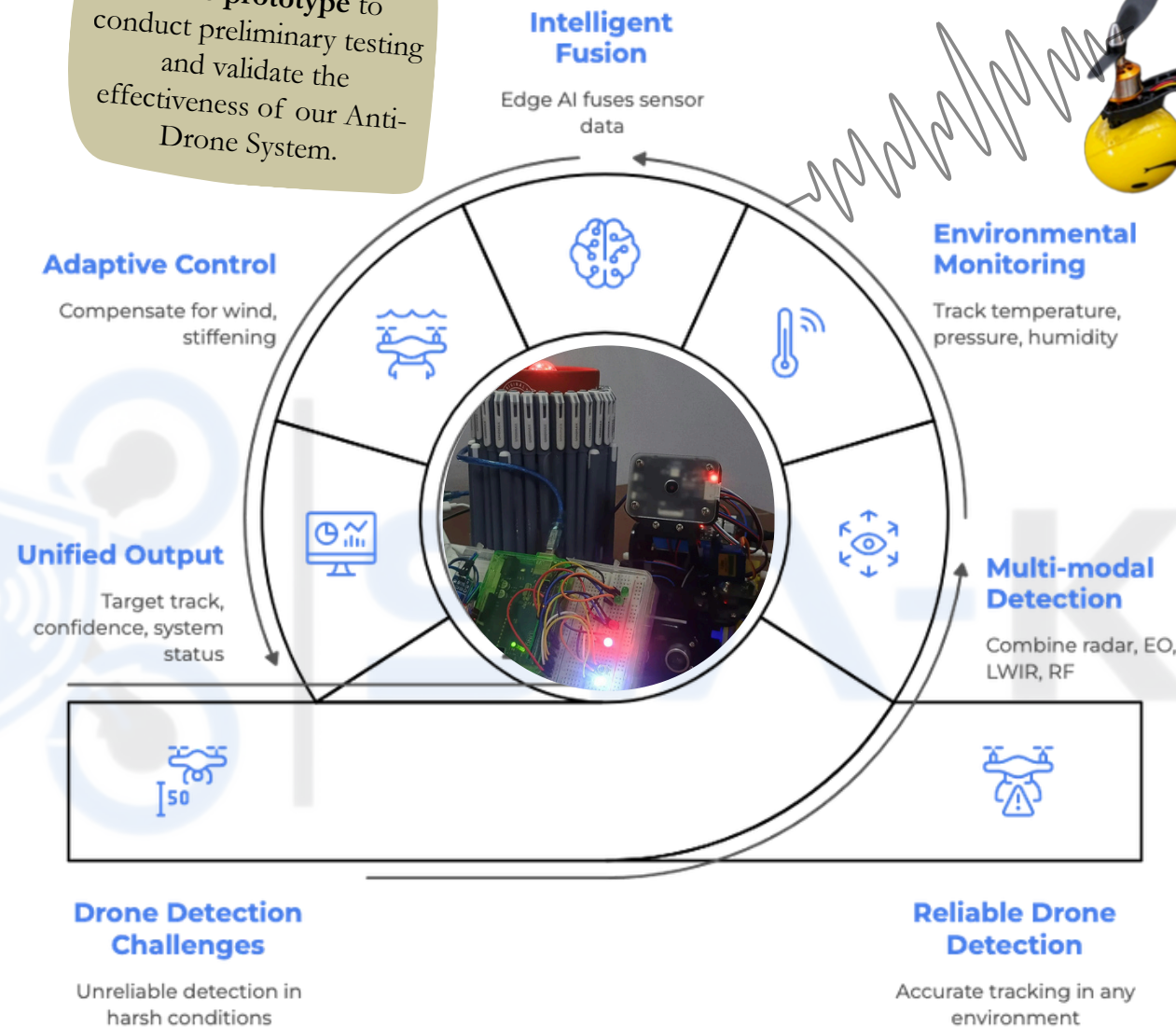
System Monitoring
Temperature • Pressure • Humidity •
Vibration • Battery • Actuators

Intelligent Processing
Edge AI • EKF • Weighted Sensor
Fusion

Adaptive Control
Real time compensation and efficient
design for wind, mechanical stiffening

Unified Output
Target Track + Confidence
Score + System Health

We developed a **custom drone prototype** to conduct preliminary testing and validate the effectiveness of our Anti-Drone System.



CLOSED LOOP ARCHITECTURE

Sense → Assess → Fuse → Track → Adapt → Correct
It detects the threat and monitors its own ability to detect it.

MULTI SENSORS SMARTER DECISIONS

Designed to withstand combined thermal, vibration and altitude stresses.

(Refer to: U.S. DoD, MIL-STD-810F, Method 520.2)

BUILT TO OVERCOME

Extreme Cold → Thermal & power-aware control
High Winds → Encoder feedback + adaptive PID
Low Pressure → Environmental & thermal zone monitoring
Fog / Snow / Dust → Reduce unreliable EO confidence
Vibration → Isolation + EKF noise correction
Connectivity Loss → Edge processing + long range telemetry

INNOVATION EDGE

Adaptive Power: Optimizes heating and power based on system health.
Failure Resilient: Continues tracking even when a sensor degrades.
Self Diagnosing: Detects degradation in sensors, batteries & actuators
Adaptive Tracking: Control parameters automatically compensate for disturbances.
Multi Modal: Radar + RF + EO + LWIR reduces single sensor dependency.
Sodium-Ion Power: Improved low-temperature resilience and safer, reliable power in high-altitude environments.

Software Architecture

AI / Algorithms

- YOLOv8 + NVIDIA TensorRT (Target detection & classification)
- EKF + Bayesian Fusion (Multi-sensor state estimation)
- Adaptive PID Gain Scheduling (Dynamic gimbal control)

C2 & Security

- React / Vue + TailwindCSS (Tactical dashboard)
- WebSockets (Real-time telemetry & radar plots)
- AES-256 (Authentication & encrypted telemetry)

Hardware Architecture

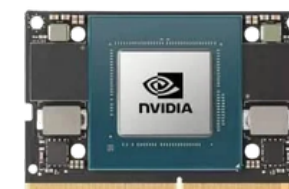
Environmental & Health

- BME280 / BMP390 (Temperature & pressure)
- Ultrasonic Wind Anemometer
- BMI088 IMU (Vibration & motion)
- INA219 / INA226 (Power telemetry)
- Multi-point Thermistors (Thermal monitoring)

Compute, Actuation & Communication

- NVIDIA Jetson Orin NX (Edge compute)
- SimpleFOC Brushless Drivers (Gimbal actuation)
- Sodium-Ion / LiFePO₄ Battery Packs
- RN2903 LoRa Module (Telemetry)

Components & Technology



Jetson-Nano Orin



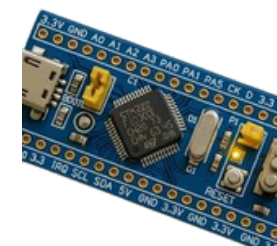
LoRa Module



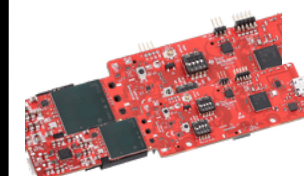
Eo Camera



FLIR Lepton 3.5 LWIR

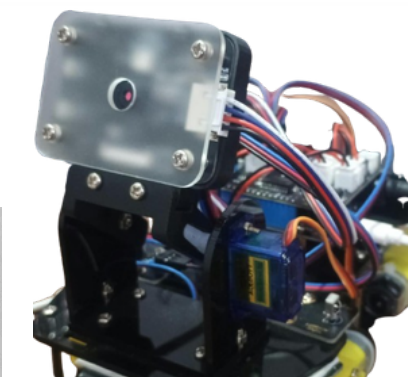
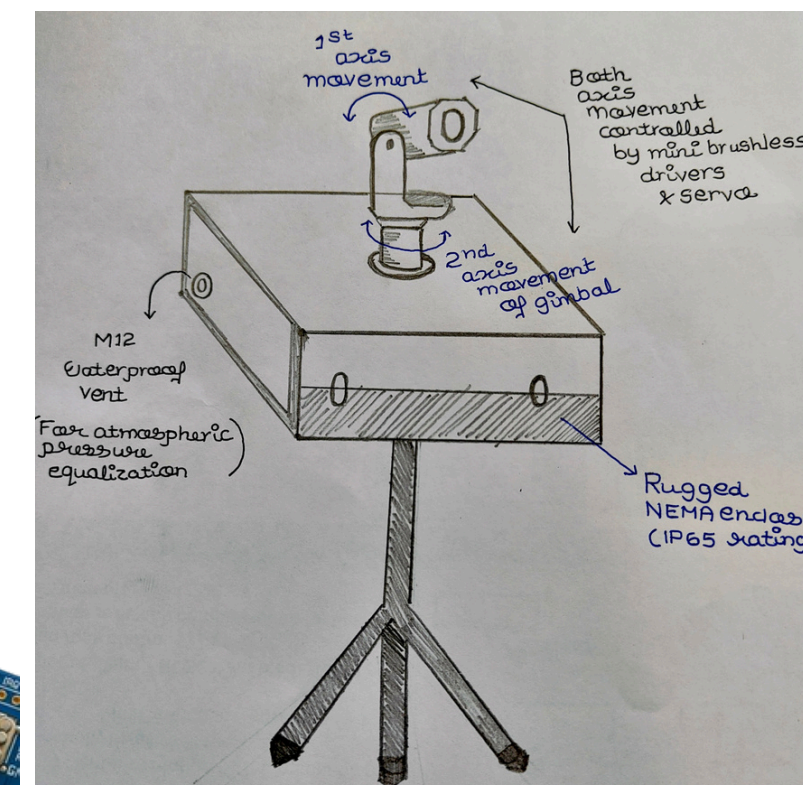


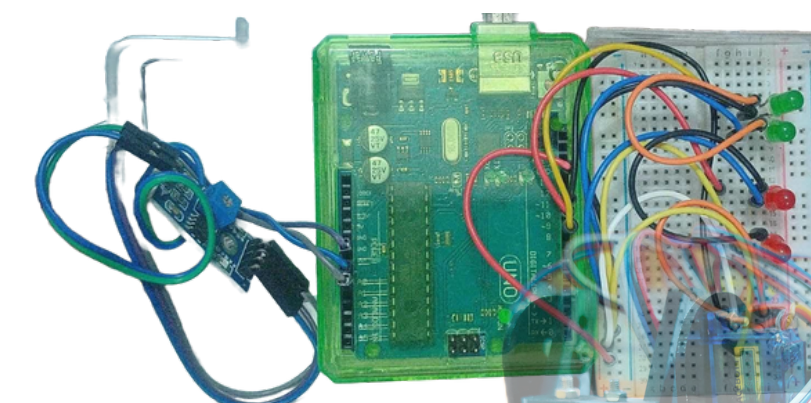
STM32 MCU


TI IWR6843
mmWave Radar


OpenCV

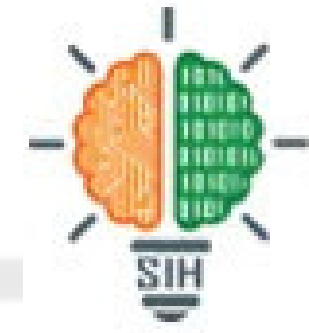
Hardware


Thermal Camera
Gimble

Signalling Unit
and Neutralizer


Main Control Unit

((For basic simulation we are currently using Arduino))



FEASIBILITY

COTS & Modular Architecture: Uses readily available COTS sensors with modular interfaces, enabling rapid prototyping, component replacement, and future upgrades

Edge AI & LoRa: Local Jetson processing and radio telemetry require zero cloud connectivity.

Closed-Loop Feedback: Software is designed to self monitor and correct mechanical lag instantly.

KEY CHALLENGES

Climate Extremes: Sub-zero battery drain and low pressure electronics overheating.

Mechanical Stress: High-altitude wind gusts and platform vibration throwing off tracking.

Sensor Blindness: Sudden fog, snow, or dust rendering optical cameras useless.

MITIGATION STRATEGY

Thermal Resilience: Sodium Ion/LiFePO4 batteries, smart BMS, and active zoned heating with Smart Enclosure

**Refer Document for outermost Enclosure R and D*

Adaptive Stabilization: Encoder feedback loops and Extended Kalman Filter (EKF) noise suppression.

Dynamic Sensor Fusion: AI dynamically re-weights sensor confidence based on real-time sensor health and environmental conditions.

Brand Collaboration
for Battery
(POWER SOURCE)

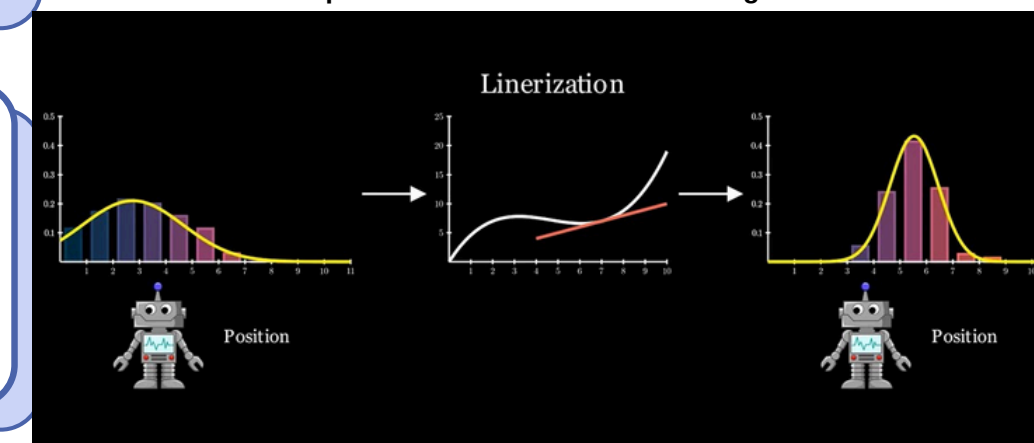


aatral esp



**Linerization Plot of ML Model
(Extended Kalman Filter)**

For removing out signal drifts and error during the workflow implimentation while actual working



SODIUM-ION
HIGH-ALTITUDE EDGE



-40°C COLD TOLERANCE
No capacity loss in severe frost



SUPERIOR SAFETY
Zero thermal runaway risk



RUGGED & DURABLE
Resists high-altitude vibration

Potential Impact on the Target Audience

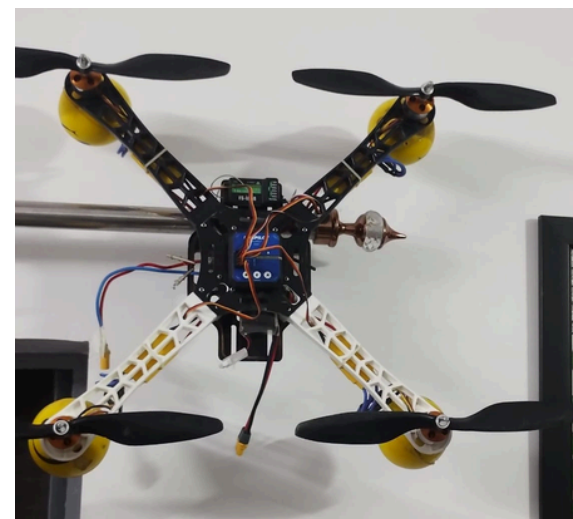
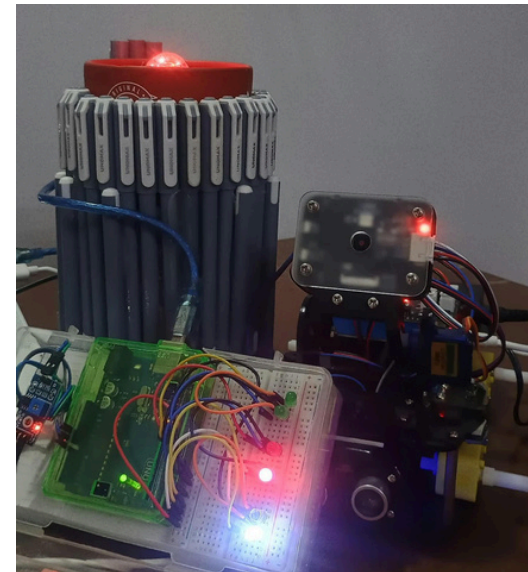
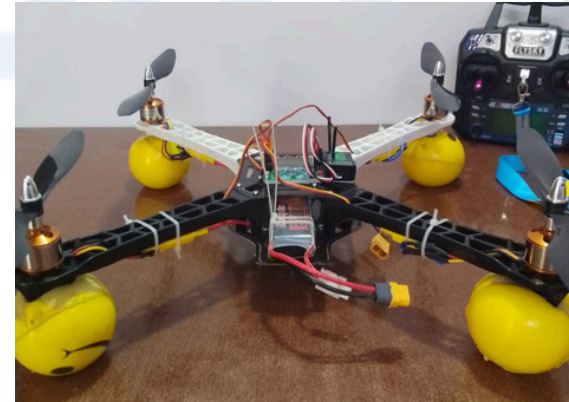
Reliable UAV tracking:
even in volatile high altitude conditions.

Real time visibility:
context aware threat assessment

Low latency Edge AI:
no dependency on external networks.

Key Outcome:
Detect - Understand - Adapt - Track

Smart Enclosure Architecture:
Temperature , Vibration and Water Resistant



Benefits of Solution

Strategic Resilience:
self-diagnosing, self-correcting & active environmental adaptation ensuring high tracking stability despite wind and vibration.

Economic Scalability:
simplifies maintenance, easy hardware upgrades, cost-effective deployment scaling without requiring full system replacements.

Future-Proof Capability:
clear expansion path from single-target interception to multi-target tracking, swarm tracking, and integration into a distributed sensor network.

Operational Reliability:
Multi-modal sensor fusion reduces the dependence on single sensor, ensuring the system continues to function effectively through changing weather and system conditions.

Bibliography

- [DRDO's Radar-Based Drone Jammer with anti drone copcage](#)
- [Ghatak UCAV Program](#)
- [DRDO Anti Drone System Overview](#)
- [Integrated Drone Detection and Interdiction System \(Mark-2\)](#)
- [U.S. DoD, MIL-STD-810F, Method 520.2, 2002.](#)

Project Deliverables

- [Research Document \(Clickable\)](#)
- [Github SiaKawach](#)
- [Outer-Enclosure R&D Work Document](#)

OUR R&D partners



MMES @ IITD
Dept. of Civil Eng

[Demonstration
Video](#)



Parameter	Existing DRDO D4 System (Baseline)	Proposed Sia-Kavach (High-Altitude Optimised)
Cost-Effectiveness	High unit cost; designed for general/plains deployment; cost not optimised for altitude-specific derating losses.	Higher component-grade cost due to industrial and military grade parts, offset by extended operational life and reduced field-failure/maintenance cost at altitude.
Sensor Suite	Phased-array radar, RF sensors, EO/IR trackers, day/night camera, laser ranging.	mm Wave FMCW radar, EO camera, LWIR thermal (FLIR Lepton 3.5), passive RF sensing array, plus a dedicated secondary Environmental & System Health sensing layer (pressure, humidity, vibration, battery health).
High-Altitude Adaptability	Not specifically documented as altitude-optimised; assumes passive ruggedisation for nominal environmental conditions.	Purpose built: active PID gain scheduling, torque-residual gimbal compensation, zoned thermal management for low pressure cooling failure, cold induced voltage- ag mitigation.
Jamming / Interception Methods	RF communication link jamming, Anti GNSS/GPS spoofing (soft kill), Laser Directed Energy Weapon (hard kill).	Passive RF sensing for early warning; architecture is focused on detection, tracking and confidence-scored classification. Kinetic/DEW hard-kill layer is outside current scope; treated as a future integration point.
SWaP-C (Size, Weight, Power, Cost)	Static tripod-mounted or vehicle-mounted configuration; designed for general field deployment.	Compact IP65 NEMA enclosed node on tripod mount; LiFePO4 power with dedicated PDB and Smart BMS; power budget explicitly allocated for active thermal management, trading some power/weight margin for altitude specific reliability.